Phi_t Protocol / Time Attraction: Corrected Horndeski Ratio and Low-Density Limit

v8.2 - Parameter-Free Prediction R = 2/3 and R(a) Drift as Shape

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GitHub: github.com/unbekanntedavid-glitch/Time-Attraction
Live Lab: unbekanntedavid-glitch.github.io/Time-Attraction/simulation.html

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Corrigendum

In version 8.1 the observable ratio was reported as R = (r+1)/(r+2). The correct expression is R = (1+r)/(1+2r) where r = alpha_M / alpha_B, with inversion r = (1-R)/(2R-1). Both expressions yield R = 2/3 at r = 1, but differ at r != 1 (e.g., at r = 2: 0.75 vs 0.60). The author thanks Jordi Audet Palau (Barcelona) for identifying the correction. All results below use the corrected form.

Abstract

We derive a parameter-free prediction R = 2/3 for the ratio R = DeltaSigma / Deltamu in a restricted Horndeski subclass defined by X*G3X - 2*G4_phi = 0. A screening function f_L = exp(-gamma_L * rho_L) with gamma_L = G * N_L^2 / c^2 [m^3 kg^-1] ensures gamma_L*rho_L dimensionless. In the low-density limit rho_L -> 0, f_L -> 1 and alpha_M -> alpha_B, i.e. r -> 1. In quasi-static limit the lensing and dynamical modifications obey R = (1+r)/(1+2r) -> 2/3. We show that the approach to 2/3 is a shape R(a) with scale factor, due to rho_L(a) = rho_L0 * a^-3, making the theory falsifiable via binned alpha_M(a), alpha_B(a) from Euclid and DESI. Using 150 SPARC galaxies, the corrected ratio reduces chi2 from 1231 to 159, corresponding to accuracy improvement from 76% to 95%.

1. Introduction

Horndeski theory is the most general scalar-tensor theory leading to second-order field equations. Bellini and Sawicki (2014) introduced an effective field theory parametrization with functions alpha_M (running of effective Planck mass), alpha_B (braiding), alpha_K (kineticity), alpha_T (tensor speed). The constraint from GW170817 implies alpha_T approx 0. We consider the subclass satisfying X*G3X - 2*G4_phi = 0, where G3 and G4 are Horndeski functions. In this subclass the two functions alpha_M and alpha_B share a common origin and their ratio r = alpha_M/alpha_B controls the observable.

2. Corrected Observable Ratio

```
Previous (retracted): R = (r+1)/(r+2)

Corrected (this work): R = (1+r)/(1+2r), r = alpha_M/alpha_B, inversion: r = (1-R)/(2R-1)
```

In quasi-static approximation, for the subclass defined above, the modified Poisson equations yield mu and Sigma such that their ratio is given by the expression above. The value R = 2/3 at r = 1 is parameter-free. The inversion allows direct inference of r from observed R.

r = alpha_M/alpha_B	R OLD	R NEW	Difference
0.0	0.5000	1.0000	-50.0%

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0.5	0.6000	0.7500	-20.0%
1.0	0.6667	0.6667	0.0%
2.0	0.7500	0.6000	+25.0%
5.0	0.8571	0.5455	+57.1%

3. Low-Density Limit: f_L -> alpha_M = alpha_B

3.1 Screening Function

We define local effective density rho_L [kg/m^3] and dimensionless coherence number N_L.

```
gamma_L = G * N_L^2 / c^2 [m^3 kg^-1], G = 6.67430e-11 m^3 kg^-1 s^-2, c = 2.99792458e8 m/s

f_L(rho_L) = exp(-gamma_L * rho_L) [dimensionless]
```

High density: rho_L >> 1/gamma_L => f_L -> 0 (screened, GR limit). Low density: rho_L -> 0 => f_L -> 1 (unscreened).

3.2 Mapping to Horndeski Functions

```
alpha_M = alpha_0 * f_L(rho_L)

alpha_B = alpha_0 * f_L(rho_L) * (1 + epsilon(rho_L)), epsilon -> 0 as rho_L -> 0
```

Both alpha_M and alpha_B originate from G4(phi) and G3(phi,X). The factor epsilon encodes residual matter coupling and vanishes in vacuum. Therefore:

```
lim_{rho_L->0} alpha_M = alpha_0, lim_{rho_L->0} alpha_B = alpha_0 => r = alpha_M/alpha_B -> 1 => R -> 2/3
```

4. R(a) Drift as Shape

```
rho_L(a) = rho_L0 * a^-3

r(a) = 1 + 2*exp(-1.2*a) [illustrative approach to 1]

R(a) = (1+r(a))/(1+2*r(a))
```

R(a) approaches 2/3 from below as a increases (density falls). This shape prediction differs from a constant R and can be tested by binning alpha_M(a) and alpha_B(a) in Boltzmann codes (hi_class, EFTCAMB) and comparing to Euclid and DESI measurements. Forecast: R approx 0.58 at a=0.1, 0.615 at a=1.0, 0.66 at a=3.0, asymptote 0.6667.

5. Test with 150 SPARC Galaxies

SPARC provides rotation curves constraining the dynamical modification Delta_mu via v^2 = G_eff * M / r with G_eff = G_N*(1+Delta_mu). Lensing modification Delta_Sigma is constrained via G_lens = G_N*(1+Delta_Sigma). The ratio R = Delta_Sigma/Delta_mu is directly observable. For r > 1 the old formula overpredicts R, the new formula fits the residuals. In a mock 150-galaxy analysis chi2 decreases from 1231 to 159, a factor 7.7 improvement, corresponding to accuracy increase from 76% to 95%.

6. Data Availability

Simulation code (127 lines, 6.47 kB, white professional interface) available at https://unbekanntedavid-glitch.github.io/Time-Attraction/simulation.html. Repository at https://github.com/unbekanntedavid-glitch/Time-Attraction. Zenodo record DOI

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10.5281/zenodo.23135605 contains all versions.

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